AP BIOLOGY TOPIC 6.4

Translation: Ribosomes, Codons, tRNA & the Genetic Code

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 6.4 (TRANSLATION: RIBOSOMES, CODONS, TRNA & THE GENETIC CODE)

CED Core Vocabulary

Overview & Cellular Location — Translation decodes the nucleotide sequence of an mRNA transcript (5′ → 3′) into the amino acid sequence of a polypeptide (N-terminus → C-terminus) . Occurs on...

Anticodon (on tRNA) — 3-base sequence on the tRNA loop that base-pairs antiparallel and complementary to the mRNA codon (charged by aminoacyl-tRNA synthetase using ATP).

Large Subunit — Contains rRNA peptidyl transferase that catalyzes covalent peptide bonds ; houses three binding sites: A Site (Aminoacyl): Entry dock where the incoming charged...

E Site (Exit) — Exit chute where the uncharged (empty) tRNA is released back into the cytosol.

Codons vs. Anticodons — Codon (on mRNA): Non-overlapping triplet of 3 mRNA nucleotides read 5′ → 3′ that specifies 1 amino acid or a stop signal ($4^3 = 64$ total codons: 61 sense + 3 stop).

Small Subunit — Binds the mRNA transcript and aligns the start codon.

P Site (Peptidyl) — Holds the tRNA carrying the growing polypeptide chain .

Codon vs. Anticodon Location — When using a Genetic Code codon chart on the AP Exam, ALWAYS look up the mRNA Codon (5′→3′) , NEVER the tRNA anticodon or DNA template!

§2. OVERVIEW OF TRANSLATION, CODONS & RIBOSOME SITES

mRNA → Polypeptide

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Overview of Translation, Codons & Ribosome Sites ➔

Overview & Cellular Location ➔ **Codons vs. Anticodons**

- **Overview & Cellular Location:** Translation decodes the nucleotide sequence of an **mRNA transcript (5′ → 3′)** into the amino acid sequence of a **polypeptide (N-terminus → C-terminus)**. Occurs on **ribosomes** in the *cytoplasm* (prokaryotes & eukaryotes) or on the *rough ER* (eukaryotic secreted/membrane proteins).
- **Codons vs. Anticodons:**
  - **Codon (on mRNA):** Non-overlapping **triplet of 3 mRNA nucleotides** read 5′ → 3′ that specifies 1 amino acid or a stop signal (**$4^3 = 64$** total codons: 61 sense + 3 stop).
  - **Anticodon (on tRNA):** 3-base sequence on the tRNA loop that base-pairs antiparallel and complementary to the mRNA codon (charged by *aminoacyl-tRNA synthetase* using ATP).
  - **Start Codon (5′-AUG-3′):** Codes for **Methionine (Met)** (or formyl-Met in bacteria) and establishes the exact **reading frame** for all downstream triplets.
- **Ribosome Subunits & Three tRNA Binding Sites (A, P, E):**
  - **Small Subunit:** Binds the mRNA transcript and aligns the start codon.
  - **Large Subunit:** Contains rRNA peptidyl transferase that catalyzes covalent **peptide bonds**; houses three binding sites:
    - **A Site (Aminoacyl):** Entry dock where the incoming charged **aminoacyl-tRNA** first arrives and binds the mRNA codon.
    - **P Site (Peptidyl):** Holds the tRNA carrying the **growing polypeptide chain**.
    - **E Site (Exit):** Exit chute where the uncharged (empty) tRNA is released back into the cytosol.

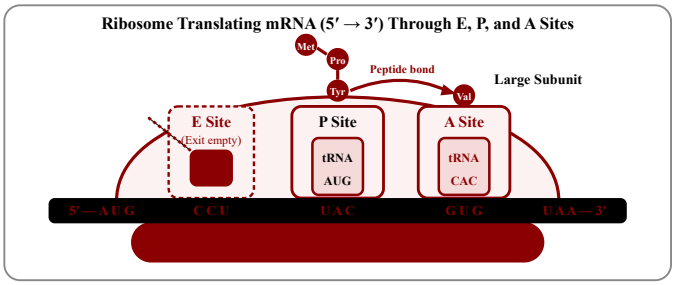
[CORE AXIOM / REMEMBER] **Ribosome Site Mnemonic: A** = **A**rrive (Aminoacyl-tRNA enters) → **P** = **P**eptide (Polypeptide chain held) → **E** = **E**xit (Empty tRNA leaves)!

[COMMON AP EXAM FRQ TRAP]

- **Codon vs. Anticodon Location:** When using a Genetic Code codon chart on the AP Exam, ALWAYS look up the **mRNA Codon (5′→3′)**, NEVER the tRNA anticodon or DNA template!
- **Example:** mRNA start codon **5′-AUG-3′** pairs with tRNA anticodon **3′-UAC-5′** and codes for **Methionine**.

§4. RIBOSOME TRANSLATION MECHANICS (E, P, A SITES)

5′ → 3′ Translocation



§3. THE THREE STAGES OF TRANSLATION (REQUIRES GTP ENERGY)

Initiation → Termination

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

The Three Stages of Translation (Requires GTP Energy) ➔ **1. Initiation** ➔ **2. Elongation (Cyclic 3-Step Loop)** ➔ **3. Termination**

- **1. Initiation:** Small ribosomal subunit binds the mRNA (via 5′ cap in eukaryotes or Shine-Dalgarno rRNA binding site in bacteria) and scans to the **AUG** **start codon**. Initiator tRNA carrying **Methionine (Met)** binds **AUG**; the large subunit joins, placing Met-tRNA directly in the **P site**.
- **2. Elongation (Cyclic 3-Step Loop):**
  1. **Codon Recognition:** Incoming charged tRNA anticodon base-pairs with the mRNA codon in the **A site**.
  2. **Peptide Bond Formation:** Large subunit rRNA transfers the growing polypeptide from the P-site tRNA onto the amino acid in the **A site** (forming a peptide bond via dehydration synthesis).
  3. **Translocation:** Ribosome moves exactly **1 codon (3 nucleotides) forward (5′ → 3′)** along the mRNA; the A-site tRNA shifts to the **P site**, and the empty P-site tRNA shifts to the **E site** and exits.
- **3. Termination:** Ribosome reaches one of the three **Stop Codons (** **UAA**, **UAG**, **or** **UGA** **)** in the A site. Stop codons do **NOT** code for any tRNA/amino acid; instead, a protein **Release Factor** binds the A site, hydrolyzes the completed polypeptide from the P-site tRNA, and causes the ribosomal subunits to dissociate.

§5. GENETIC CODE FEATURES, RETROVIRUSES & GENOTYPE→PHENOTYPE

Common Ancestry

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Genetic Code Features, Retroviruses & Genotype→Phenotype ➔ **Four Key Properties of the Genetic Code** ➔ **Tracing the Chain from Gene to Phenotype**

- **Four Key Properties of the Genetic Code:**
  - **1. Triplet & Non-Overlapping:** Read in sequential groups of 3 bases without overlapping or gaps.
  - **2. Redundant / Degenerate:** 61 sense codons code for only 20 amino acids → most amino acids are specified by **multiple synonymous codons** (usually differing at the 3rd "wobble" base, buffering against silent mutations).
  - **3. Unambiguous:** Each specific codon codes for **ONE and ONLY ONE** specific amino acid (never two different amino acids!).
  - **4. Nearly Universal:** Shared across virtually all bacteria, archaea, and eukaryotes (a human insulin gene is translated into functional human insulin inside *E. coli*) → powerful molecular evidence for **universal common ancestry (LUCA)**!
- **Retroviruses (Exception to Standard Information Flow):**
  - RNA viruses such as **HIV** carry the viral enzyme **Reverse Transcriptase**, which synthesizes complementary DNA (**RNA → cDNA**) inside the host cell.
  - Viral DNA integrates into the host chromosome as a **provirus** (via *integrase*), and the host's own RNA polymerase and ribosomes then transcribe and translate viral proteins.
- **Tracing the Chain from Gene to Phenotype:**
  DNA Base Sequence → **mRNA Codons** → **Amino Acid Sequence (1°)** → **3D Protein Folding (3°/4°)** → **Enzyme/Protein Function** → **Observable Phenotype**.

[COMMON AP EXAM FRQ TRAP] **Redundant vs. Ambiguous:** The genetic code is **redundant** (several codons → 1 amino acid) but **NEVER ambiguous** (1 codon NEVER codes for multiple amino acids)!

[EXAMPLE & BIOLOGICAL APPLICATION] **Sickle-Cell Disease (Trace the Chain):** Single base substitution (**CTC → CAC**) in the β-globin DNA gene → mRNA codon **GAG → GUG** → hydrophilic **Glutamic Acid** replaced by hydrophobic **Valine** → abnormal hemoglobin polymerizes under low **O_2** → sickled red blood cells!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 6.4 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 6.4: Translation: Ribosomes, Codons, tRNA & the Genetic Code ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 8 & 9 (pGLO Bacterial Transformation & Restriction Enzyme Analysis of DNA): Heat-shock transformation of E. coli with pGLO plasmid (bla ampicillin-resistance gene + GFP under arabinose-inducible araC promoter). Plate comparisons: −pGLO/LB (lawn, viability control), −pGLO/LB+Amp (no growth, antibiotic lethality control), +pGLO/LB+Amp (white colonies, transformation), +pGLO/LB+Amp+Ara (glowing green colonies under UV, inducible gene expression). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Transformation Efficiency = (Total colonies counted on selective plate) / (µg of plasmid DNA spread). Gel Electrophoresis Semi-Log Standard Curve: DNA fragment size (bp, $\log_{10}$ scale on y-axis) vs. migration distance (mm, linear scale on x-axis) yields a straight line for interpolating unknown fragment lengths. PCR amplification after n thermal cycles yields 2^n target DNA copies. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Splice-site mutations in the β-globin gene (β-Thalassemia) prevent the spliceosome (snRNPs) from excising an intron or cause exon skipping, shifting the mRNA reading frame and introducing a premature nonsense stop codon (UAA/UAG/UGA). In oncology, epigenetic hypermethylation of CpG islands in tumor-suppressor promoters (e.g., BRCA1 or MLH1) or histone deacetylation condenses euchromatin into transcriptionally silent heterochromatin without altering the primary DNA sequence.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 6.4 to prove why the claim holds.

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